Examiner only

9. (a) Over 180 million tonnes of ammonia are manufactured each year.

The main use of ammonia is in the production of salts such as ammonium sulfate, which is used as a fertiliser.

$$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

Explain why this is an acid-base reaction. [1]

....

....

(b) Sodium hydroxide reacts with ammonium sulfate to form ammonia, sodium sulfate and water as shown in the equation below.

$$(NH_4)_2SO_4 + 2NaOH \longrightarrow 2NH_3 + Na_2SO_4 + 2H_2O$$

A 1.86 g sample of ammonium sulfate was neutralised by exactly 26.70 cm^3 of a sodium hydroxide solution.

Calculate the concentration, in $mol\,dm^{-3}$, of the sodium hydroxide solution used. [3]

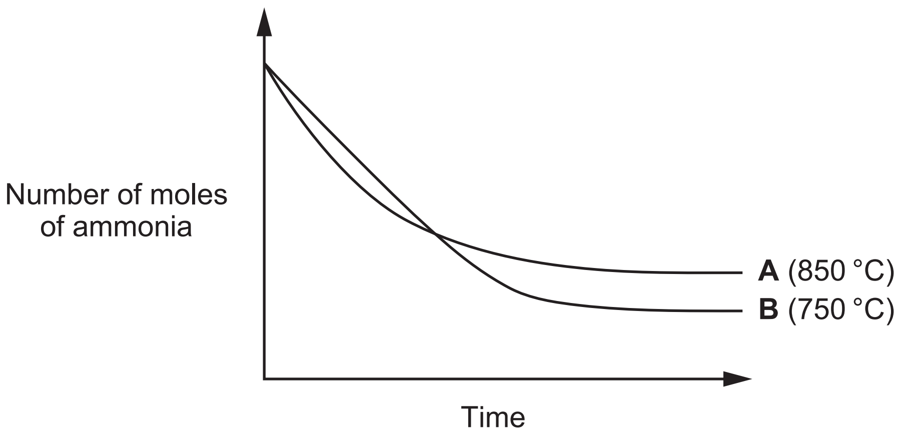
Concentration = ... $mol\,dm^{-3}$

Examiner only

(c) Another use of ammonia is in the production of nitric acid. In the first part of this process ammonia is oxidised in air.

$$4NH_3(g) \quad + \quad 5O_2(g) \rightleftharpoons 4NO(g) \quad + \quad 6H_2O(g)$$

The graph below shows how the number of moles of ammonia present changes as the reaction proceeds until equilibrium is reached.



(i) State Le Chatelier's principle. [1]

....

....

(ii) A student looked at curves **A** and **B** and said that the forward reaction is exothermic.

Is he correct? Justify your answer by using Le Chatelier's principle. [2]

....

....

....

....

(iii) On the graph, draw a curve that represents the reaction at 850 °C but with a catalyst added to the reaction mixture.

Label this curve **C**. Explain the shape of the curve. [2]

....

....

....

2410U101
11

Examiner only

(d) A solution of nitric acid has a concentration of 0.0550 mol dm^{-3}.

Calculate its pH. [1]

pH = ...

(e) A student said that the bonds in an ammonia molecule are not purely covalent.

Explain why she is correct. [2]

....

....

....

....

12